How-To Library

Informatica

Performance Tuning and Sizing Guidelines for Amazon Redshift Connector

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Abstract

When you use Amazon Redshift Connector, multiple factors such as data set size, hardware parameters, and mapping parameters, impact the connector performance. You can optimize the performance by analyzing your data set size, using the recommended hardware, and tuning these parameters appropriately. This article describes general reference guidelines to help you analyze the data set and tune the performance of Amazon Redshift Connector.

Supported Versions

- Informatica Cloud® Amazon Redshift Connector Fall 2017

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Overview

Performance tuning is an iterative process where you analyze the performance, estimate and define parameters that impact the performance, monitor, and then adjust the results as required. Sizing is a process where you analyze the data set and use the recommended hardware to improve the performance.

This document describes the key data size, hardware, and Informatica mapping parameters that you can tune to optimize the performance of Amazon Redshift Connector.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Amazon Redshift Connector might vary based on individual environments and other parameters even when you use the same data.

Performance Sizing Recommendations

You can tune the hardware parameters based on the data set and use the recommended hardware capacities to optimize the performance of the machine where the Secure Agent runs.

Performance sizing recommendations depend on the size of the data set. Consider the sizing recommendations for the following types of data set:

Small data set

A data set is considered to be a small data set if you read from or write to less than 10 GB of data.

For a small data set, you can read 10 GB of data from an Amazon Redshift source to an EC2 file system in `0:30:00(HH:MM:SS)` minutes and write data to an Amazon Redshift target in `0:14:00(HH:MM:SS)` minutes.

The following table describes the sizing recommendations for the system that processes small data sets:

Requirement	Sizing Recommendation
VCPUs	2
Memory	8 GB
Storage	100 GB

Medium data set

A data set is considered to be a medium data set if you read from or write to less than 200 GB of data.

For a medium data set, you can read 200 GB of data from an Amazon Redshift source to an EC2 file system in `09:30:00(HH:MM:SS)` hours and write data to an Amazon Redshift target in `04:15:00(HH:MM:SS)` hours.

When you write data to a target, the Secure Agent creates two partitions.

The following table describes the sizing recommendations for the system that processes medium data sets:

Requirement	Sizing Recommendation
VCPUs	8
Memory	8 GB
Storage	500 GB

Large data set

A data set is considered to be a large data set if you read from or write to less than 1 TGB of data.

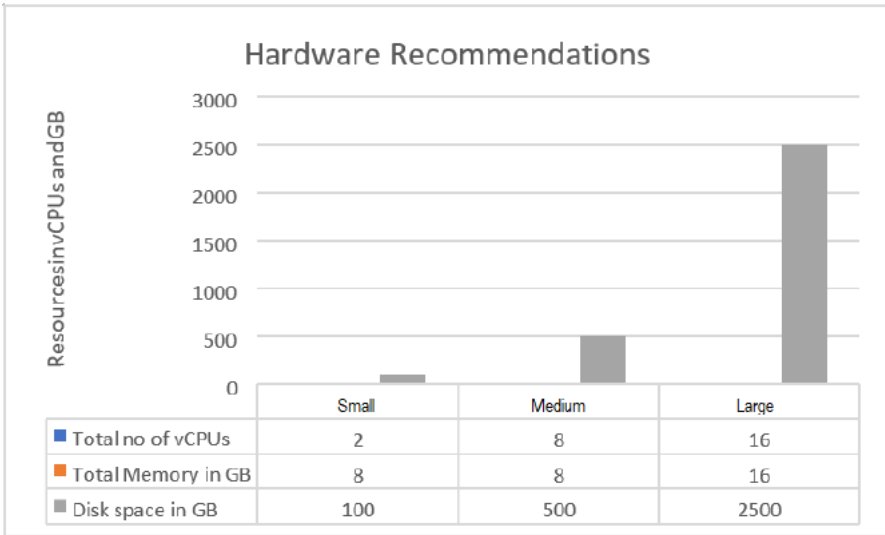
For a large data set, you can read 1 TB of data from an Amazon Redshift source to an EC2 file system in `24:00:00(HH:MM:SS)` hours and write data to an Amazon Redshift target in `10:30:00(HH:MM:SS)` hours.

When you write data to a target, the Secure Agent creates four partitions.

The following table describes the sizing recommendations for the system that processes large data sets:

Requirement	Sizing Recommendation
VCPUs	16
Memory	16 GB
Storage	2.5 TB

The following image represents the hardware requirements along with the change in data set and change in individual component settings:



Performance Tuning Parameters

You can optimize the performance of Amazon Redshift mappings by tuning the following parameters:

- Amazon Redshift mapping
- Amazon Redshift cluster properties

Tune the Mapping

You can tune the following properties in the mapping to optimize the performance:

Buffer block size

The buffer block size determines the size of the blocks of data used to move rows of data from the source to the target. You can specify the buffer block size in bytes or as percentage of total memory.

DTM buffer size

The DTM buffer size determines the size of the buffer blocks that the Secure Agent uses to move data from sources to targets. All the buffer blocks are allocated from the DTM buffer pool.

Batch size

Number of records that Secure Agent writes at a time while loading data.

Enable compression

Enables compression to compress the staged files before writing the files to Amazon Redshift.

The mapping performance improves when the Secure Agent compresses the staged files.

The following table lists the tuning recommendations for the mapping parameters according to the size of the data set:

Mapping Properties	Small Data Set	Medium Data Set	Large Data Set
Buffer Block Size	5 MB or minimum row length	5 MB or minimum row length	5 MB or minimum row length
DTM Buffer Size	50 MB	100 MB	200 MB

Mapping Properties	Small Data Set	Medium Data Set	Large Data Set
Batch Size	20 million	200 million	2 billion
Enable Compression	Yes	Yes	Yes

Tune the Amazon Redshift Cluster Properties

You can tune the following properties at the mapping level for the Amazon Redshift cluster:

Properties	Small Data Set	Medium Data Set	Large Data Set
Redshift Compute Node Type	dc1.8xlarge	dc1.8xlarge	dc1.8xlarge
Number of Redshift Compute Nodes	2	2	4

Note: When you enable partitioning for a mapping to read from Amazon Redshift, the UNLOAD command runs quickly in concurrent execution than sequential execution in the cluster.

Optimizing the Amazon Redshift Mapping

You can configure the following parameters to optimize the Amazon Redshift mapping performance:

Instance size for Informatica Secure Agent machine on EC2 system

When you provision an Amazon Redshift EC2 instance, you must choose the instance size of your nodes appropriately as some workloads are CPU intensive while others are disk I/O or memory intensive.

Use the following guidelines to choose the right instance size for the Informatica Secure Agent machine on EC2 system:

- For memory intensive jobs, you must ensure that the `m2` family instance size that you intend to use have enough memory and CPU power to perform the task.
- For CPU intensive jobs, you must ensure that you have a `c2-series` instance.
- If the jobs are both memory and CPU intensive, you must ensure that you use the `cg1.4xlarge` or `cc2.8xlarge` instance as they have suitable memory and CPU power to handle a heavy workload.

Amazon Redshift cluster

When you provision an Amazon Redshift cluster, you must choose the **Redshift Compute Node Type** and **Number of Redshift Compute Nodes** parameters appropriately.

Note: Choose the Amazon Redshift cluster in the same region where the Secure Agent machine runs.

CPU clock speed

If your mappings consist of multiple CPU bound transformations in a medium or large data sets, Informatica recommends that you use high clock speed processor.

Network

If you need to shuffle or sort a large amount data, you must use the `cg1.4xlarge` and `cc2.8xlarge` instances with 10 Gb of network speed. When the data is in memory, multiple applications are network bound.

Note: You can see how much data shuffles or sorts across the network from the **UI Monitoring Application**.

Local flat file staging

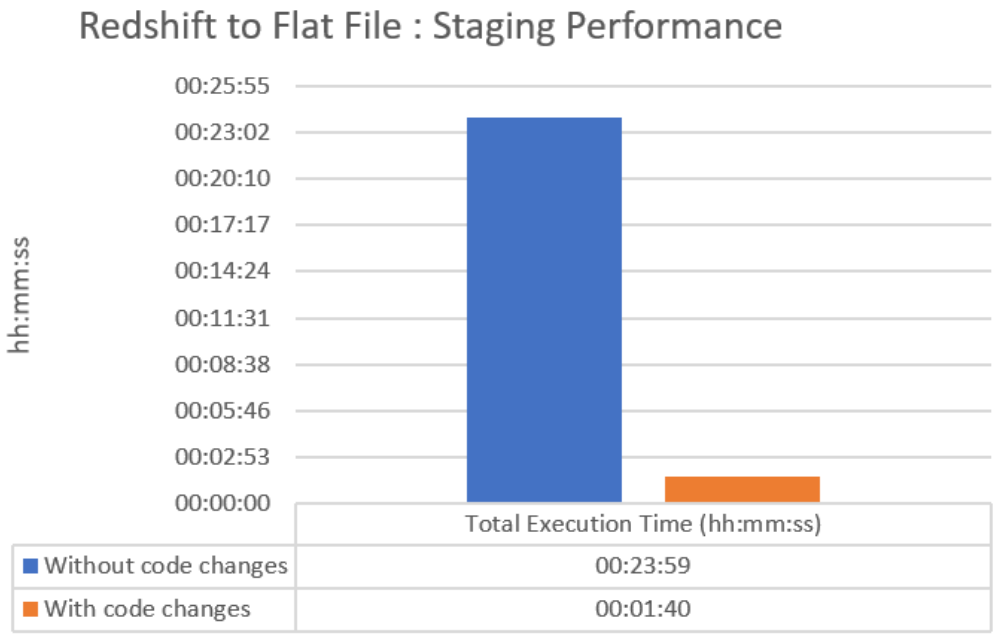
Informatica has redesigned the way Data Integration stages the data when you read from and write to an Amazon Redshift target. This has significantly improved the performance in the local staging execution time.

You can optimize the mapping performance by setting the staging property in the Secure Agent to create a flat file in a temporary folder to stage the data locally before reading from or writing to Amazon Redshift.

Reading from an Amazon Redshift source

When you set the staging property `INFA_DTM_RDR_STAGING_ENABLED_CONNECTORS` for the Tomcat to the plugin ID of the Amazon Redshift Connector in the Secure Agent properties, Data Integration, by default, creates a flat file locally to stage the data and then reads the data from the staging file from the Amazon Redshift source. For more information on how to set the property, see the help for Amazon Redshift Connector.

The following graph shows the performance comparison of the execution time before and after setting the staging property for a mapping that reads from an Amazon Redshift source and writes to a flat file target:



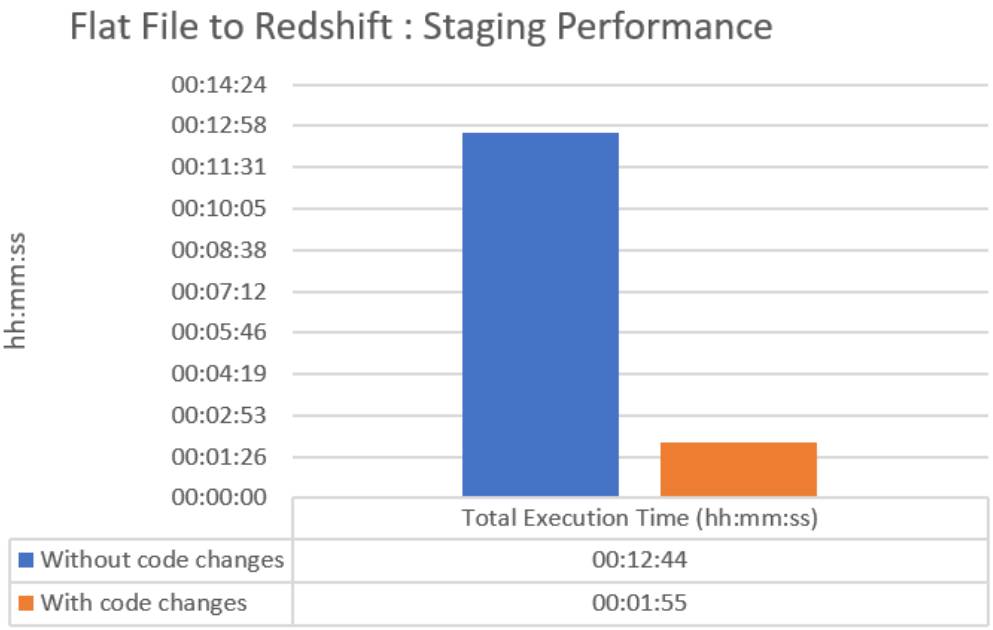
Writing to an Amazon Redshift target

When you set the staging property `INFA_DTM_STAGING_ENABLED_CONNECTORS` for the Tomcat to the plugin ID of the Amazon Redshift Connector in the Secure Agent properties, Data Integration, by default, creates a flat file locally to stage the data and then loads the data from the staging file to the Amazon Redshift target.

You can find the plugin ID in the manifest file located in the following directory: `<Secure Agent installation directory>/downloads/<Amazon Redshift package>/CCIManifest`.

For more information on how to set the property, see the help for Amazon Redshift Connector.

The following graph shows the performance comparison of the execution time before and after setting the staging property for a mapping that reads from a flat file source and writes to an Amazon Redshift target:



Informatica recommends that you add three processor cores to the agent machine to increase the processing power of the mapping.

The following table shows the resource sizing in the old and new environment for a mapping that reads from a flat file source and writes to an Amazon Redshift target:

Staging	CPU Requirement (cores)	Memory Requirement (GB)
Without code changes	3	2 GB (JVMHeap - 1 GB and DTM Buffer Size 1 GB)
With code changes	3	2 GB (JVMHeap - 1 GB and DTM Buffer Size 1 GB)

When you use three cores, the performance is optimized and the local staging execution time is far lesser. Though Informatica recommends that you include three cores for an Amazon Redshift mapping execution, it is not mandatory. However, when you include lesser cores, the mapping performance is impacted but it does not result in execution failures.

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